Manisha Muduli

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Google Scholar: scholar.google.com/citations?user=A17pXQsAAAAJ
Portfolio: manisha-muduli.github.io/research-portfolio/

Summary

Photonics Scientist with a Ph.D. in Electrical and Computer Engineering and 7+ years of experience in semiconductor nanofabrication, thin-film processing, nanomaterials synthesis, and device characterization. Research spans III–V/silicon integration, infrared photodetectors, and process–structure–performance relationships, with experience in shared cleanroom user support, fabrication-course development, multi-institutional research, and 300 mm wafer processing.

Education

The Ohio State University Aug 2026
Ph.D., Electrical and Computer Engineering
Dissertation: Heterogeneous Integration of Antimonide-Semiconductors with Silicon for Short-Wave Infrared Photodetectors.
Advisor: Prof. Sanjay Krishna.

The Ohio State University Dec 2025
M.S., Electrical and Computer Engineering

University of Pennsylvania May 2020
M.S.E., Nanotechnology
Thesis: Synthesis of Size Tunable InAs Quantum Dots for the Fabrication of SWIR Photodiode.
Advisor: Prof. Cherie Kagan.

Amity University, Noida May 2018
B.Tech., Nanotechnology
Thesis: Investigation on the Nucleation and Growth of Palladium Nanoparticles by Electrodeposition.
Advisor: Prof. Sankara Sharma V Tatiparti.

Research and Professional Experience

QC82 Inc. Aug 2026 – Present
Photonics Scientist · Charlottesville, VA

- Design and simulate single-photon detectors targeting high efficiency and low dark-count rates.
- Develop and optimize fabrication processes for InGaAs-based SPADs targeting 1550 nm sensing applications.
- Interface with clients and external vendors on process design, integration requirements, and technical deliverables.
- Establish device-characterization capabilities through equipment selection and measurement-system planning.

Intel Corporation May – Aug 2025
Research Intern, 3D Semiconductor Integration Technology · Hillsboro, OR

- Executed 300 mm wafer bonding & layer transfer processes in a Class 1 cleanroom for 3D heterogeneous integration.
- Characterized and analyzed bond integrity via defect inspection, interface analysis, and bond energy measurements.
- Supported Intel–supplier discussions by analyzing & organizing technical data, driving improvements in supplier tools.
- Presented data-driven results to Intel research leadership, contributing to next-phase development strategy.

The Ohio State University May 2021 – Aug 2026
Graduate Research Associate / Research Assistant · Columbus, OH

- Developed III–V/Si heterogeneous integration flows using transfer printing and bonding for SWIR photonic devices.
- Developed lithography, thin-film deposition, etching, and metallization processes for III–V/Si integration.
- Applied DOE, TCAD, and materials characterization for process optimization and device analysis.
- Coordinated cross-functional research and proposal development with internal and external collaborators.

Research and Professional Experience Continued

Singh Center for Nanotechnology, University of Pennsylvania Jun 2019 – Apr 2021

Graduate Student Fellow · Philadelphia, PA

- Executed microfabrication processes and thin-film metrology at the Quattrone Nanofabrication Facility.
- Developed process travelers for ESE 336 and ESE 536 and optimized a solar-cell fabrication module for ESE 336.
- Served as an assistant instructor for the August 2019 QNF Bootcamp, providing hands-on cleanroom training.

Kagan Lab, University of Pennsylvania Jan 2019 – May 2020

Research Assistant · Philadelphia, PA

- Researched size-tunable III–V quantum dots under Prof. Cherie Kagan for SWIR photodetectors.
- Developed quantum-dot synthesis recipes and fabricated spin-coated InAs-on-Si photodetectors; characterized materials with UV–Vis spectroscopy and devices with IV measurements.

Visiting Research Appointments

Tyndall National Institute Dec 2024

Visiting Researcher · Cork, Ireland

- Collaborated with Prof. Brian Corbett and Prof. Fatima Gunning on GaAsSb membrane transfer for SWIR silicon photonics, gaining experience with X-Celeprint transfer printing and applying the techniques at Ohio State.
- Exchanged research insights on III–V transfer onto Si and SOI, including InGaAsSb integration for MWIR applications.

University of Wisconsin–Madison Dec 2022

Visiting Researcher · Wisconsin, United States

- Collaborated with Prof. Zhenqiang Ma's group on semiconductor membrane-transfer approaches, transferring Si and GaAs membranes onto silicon and implementing the techniques at Ohio State.
- Studied material-specific processing requirements and collaborated on silicon membrane transfer onto GaAsSb.

Earlier Research Experience and Technical Training

Indian Institute of Technology Bombay Feb – Apr 2018

Research Assistant · Mumbai, India

- Modeled nucleation and growth during electrodeposition of non-spherical palladium particles for the bachelor's thesis.
- Modified Scharifker–Hills models to represent instantaneous and progressive nucleation processes.

Amity University, Noida May – Jun 2017

Research Intern · Noida, India

- Investigated the effect of levetiracetam on copper nanoparticles deposited on FTO-coated glass using cyclic voltammetry.
- Synthesized copper nanoparticles and characterized them with dynamic light scattering and UV–Vis spectroscopy.

i3indya Technologies Jun – Jul 2016

Embedded Systems and IoT with Arduino | Mumbai Metropolitan Region

- Completed practical training in embedded systems and Internet of Things using Arduino. Coursework covered C and embedded C programming, microcontroller interfacing, serial communication, sensors and actuators, and connected-device applications.

Research Funding and Collaborative Leadership

- Applied for and received **$15,000** through TNI–OSU/IMR Catalyst Funding 2024 (Grant No. GR137976) to support research initiatives.
- Led and managed multi-institutional research projects through Intel CAFE (Grant No. GR129846) and OSU–TNI–IMR; coordinated collaborators and supported proposal development.

Research Projects

Doctoral Research

GaAsSb/Si SACM Avalanche Photodiodes for 1.55 µm Detection Jul 2024 – Aug 2026

- To evaluate band alignment in GaAsSb/Si photodiodes, used X-ray photoelectron spectroscopy (XPS) to determine the heterojunction band offsets, identifying a conduction-band offset below 0.1 eV.
- Designed and optimized separate absorption, charge, and multiplication (SACM) structures using Silvaco TCAD.
  Tuned the absorber, charge, and multiplication layers to control electric-field distribution, carrier transport, and device behavior.
- Fabricated GaAsSb/Si devices and used current–voltage and capacitance–voltage measurements to evaluate junction behavior, breakdown characteristics, and process-related variability.
  Correlated electrical behavior with interface chemistry and processing conditions.

Heterogeneous Integration of III–V Materials with Silicon for SWIR Photodetectors Dec 2023 – Apr 2025

- Developed transfer-printing and fabrication workflows to integrate III–V absorbers with silicon, then used dark and illuminated current–voltage and capacitance–voltage measurements to demonstrate an electrically active GaAsSb/Si interface and photoresponse at 1.55 µm.
- Formulated membrane-release, transfer, and bonding methods for InGaAs, GaAsSb, and InGaAs/GaAsSb superlattices within silicon-integrated photodetector fabrication workflows.
- Simulated and fabricated n–i–p, p–i–n, and SACM structures, integrating lithography, thin-film deposition, etching, and metallization to create and evaluate heterogeneous junctions.

Selective Etchant Studies for III–V Materials Dec 2023 – Mar 2024

- Investigated selective wet etchants for III–V materials to enable controlled removal of underlying layers while preserving the materials required for heterogeneous integration.
- Evaluated etch selectivity, etch rate, surface morphology, and process compatibility to optimize material removal and prepare III–V layers for transfer-printing workflows.

Zn Diffusion in InGaAs/GaAsSb Superlattices Using ALD Deposited ZnO Apr 2022 – Feb 2024

- Investigated ALD-deposited ZnO as a dopant source for planar InGaAs/GaAsSb photodiodes by varying film thickness, diffusion temperature, and time.
  SIMS and current–voltage/capacitance–voltage measurements confirmed p–n junction formation and identified shallow vertical diffusion with comparatively wider lateral diffusion.
- Correlated secondary-ion mass spectrometry (SIMS) dopant profiles with diode characteristics to evaluate zinc incorporation and junction depth.
  Used the observed vertical and lateral diffusion behavior to define process limitations and the need for further optimization.

Design of InGaAsSb MWIR Detectors for Integration onto Silicon Aug 2025 – Feb 2026

- Studied and designed InGaAsSb-based p–i–n (PIN) and nBn detectors lattice-matched to GaSb substrates for detection at 3 µm, targeting integration onto silicon.
- Investigated sacrificial-layer materials and structures and different epitaxial stacks to support selective etching and release of InGaAsSb detector membranes for micro-transfer printing.

Master's Research Projects

Fabrication and Optimization of Silicon Solar Cells Jun 2019 – Feb 2021

- Investigated busbar and finger geometry to optimize silicon solar-cell performance, using fabrication and electrical characterization results to refine device layouts and achieve a maximum efficiency of 12%.
- Varied the number of busbars and fingers and the finger width to examine the balance between electrical collection and optical shading.
- Designed and implemented a printed circuit board for mounting and testing solar cells. Developed and standardized parameterized fabrication recipes and incorporated the process into the ESE 336 laboratory course.

Research Projects Continued

Simulation of Silicon Solar Cells Using TCAD Aug 2020 – Mar 2021

- Used NovaTCAD to design and simulate two-dimensional silicon solar cells and investigate the effects of doping, metal contacts, busbar dimensions, finger dimensions, and window geometry on device performance.
- Examined the effect of contact geometry and optical shading on current–voltage characteristics, using the simulations to interpret the tradeoffs involved in solar-cell design.

Nanomaterials and Nanofabrication

Silicon Dioxide Thin Film Properties During Wet Thermal Oxidation Aug 2020 – Mar 2021

- Investigated silicon dioxide films grown by thermal oxidation, comparing dry- and wet-oxidation processing and examining the resulting film uniformity, stress, thickness, and refractive index.
- Used thin-film metrology to relate oxidation conditions to the physical and optical properties of the oxide films.

Fabrication and Characterization of FinFETs Aug 2020 – Mar 2021

- Fabricated finFET features on silicon substrates using electron-beam lithography and optimized the lithography process for pattern definition.
- Combined electron-beam patterning with metal deposition and lift-off as part of the device-fabrication workflow.

Synthesis of Size Tunable InAs Quantum Dots for SWIR Photodiodes Jan 2019 – May 2020

- Investigated synthesis conditions for size-tunable SWIR InAs quantum dots by varying precursor and reducing-agent ratios and characterizing optical absorption. Synthesized dots up to 7.02 nm in diameter with absorption at 1530 nm.
- Used InAs quantum dots absorbing at 1470 nm to fabricate infrared photodiodes with external quantum efficiency up to 4.1%. Spin-coated quantum dots onto silicon, varied absorber thickness, and used current–voltage measurements to study device behavior and photoresponse at 1300–1470 nm.
- Conducted this master's thesis research under Prof. Cherie Kagan. Developed synthesis approaches for larger InAs, InAsSb, and InSb quantum dots to extend infrared absorption, using UV–visible spectroscopy to assess material quality and bandgap tunability.

Thin Dry Silicon Oxide Films Grown by Thermal Oxidation Jun 2019 – Jan 2020

- Characterized silicon oxidation across 900–1050 degrees Celsius and processing times of 10–200 minutes using ellipsometry, reflectometry, and growth-model analysis. Thickness mapping identified measurement limitations for approximately 30 nm films and lower measured variation for films thicker than 50 nm.
- Performed oxidation in the Sandvik fused-silica tube furnace at the Quattrone Nanofabrication Facility. Measured thickness with a Woollam ellipsometer and Filmetrics F50 reflectometer, and extracted oxidation-rate constants and activation energies using the Deal–Grove model and the Gerlach–Maser–Saad method.
- Assessed film uniformity using thickness mapping, standard deviation, and coefficient of variation to distinguish measured variation from limitations of thin-film metrology.

Stress and Refractive Index in Silicon Oxide Films Grown by Dry Thermal Oxidation 2020

- Investigated thickness-dependent optical properties of silicon oxide grown at 900–1050 degrees Celsius using ellipsometry, observing a refractive-index decrease from approximately 1.78 to 1.46 as film thickness increased.
- Compared stress estimates from Stoney's and Goklaney's equations with KLA Tencor P7 profilometer measurements. Identified measurement inconsistency and film nonuniformity as limitations in interpreting stress and film density.

Engineering Innovation

Three Dimensional Printed Braille Generator 2022 – 2024

- Collaborated in a team of four to design and demonstrate a 3D-printed Braille generator, earning first place at the 2022 Make OHI/O makeathon.
- Designed the 3D-printable model in Tinkercad, accounting for minimum printable feature dimensions and aspect-ratio constraints to accommodate features at Braille scale.
- Continued development of the device and demonstrated the project at Show OHI/O in April 2024.

Course Projects

Two Dimensional Graphene Sensor for pH Sensing Using Microfluidics Jan – May 2019

- Fabricated a graphene transistor on a silicon chip through bilayer-resist lithography, titanium/gold contact deposition, and metal lift-off. Used electrochemical transfer to place a graphene monolayer onto the patterned device and oxygen-plasma etching to define graphene channels.
- Fabricated a PDMS microfluidic cap using soft lithography, integrated it with the graphene device, and mounted the assembly on a printed circuit board.
- Measured current–voltage behavior using two- and three-probe methods, examined channel-geometry effects, and performed transmission-line and Kelvin measurements. Tested the electrical response under dry conditions and after passing buffer solutions through the microfluidic channels.

Directed Self Assembly of a Block Copolymer in a Research Setting Jan – May 2019

- Investigated directed block-copolymer assembly in silicon trenches and holes using electron-beam lithography, reactive ion etching, polymer deposition, and solvent-vapor annealing. SEM and AFM revealed uneven assembly and incomplete blocking-polymer removal, identifying trench geometry, etching, and annealing conditions for further investigation.
- Created CAD-designed trench and hole patterns to guide self-assembly of a polystyrene–polydimethylsiloxane block copolymer. Spin-coated the polymer, performed toluene solvent-vapor annealing, and used sequential CF4 and oxygen plasma etching to expose the structures.
- Used reflectometry and ellipsometry to characterize resist and polymer thickness, and SEM and AFM to compare assembly across different grating geometries.

Fabrication and Testing of an Optoelectronic Device Using CdSe Quantum Dots Jan – May 2019

- Fabricated CdSe quantum-dot devices using ligand exchange and evaporated indium/gold contacts, then used two- and four-point probe measurements to assess resistance and photoconductivity. Observed a small photocurrent, while the expected transistor behavior was not resolved.
- Synthesized quantum dots at different reaction times and characterized size-dependent optical response using absorption and emission spectroscopy, comparing spectral shifts with observed solution color.
- Spin-coated quantum dots onto oxidized silicon, exchanged long surface ligands for shorter thiocyanate ligands, and formed contacts by thermal evaporation through shadow masks followed by nitrogen annealing.

SOI Fabrication of Comb Drive Actuators and Cantilever Beams Jan – May 2019

- Designed and fabricated SOI comb-drive actuators and cantilever beams with varied beam widths and comb gaps to investigate electromechanical behavior. Compared measurements with theoretical calculations, recording vertical cantilever deflection of approximately 0.35 µm below 60 V and resonance on the order of 10 kHz.
- Used Layout Editor to design structures for investigating the relationship between electrostatic force and mechanical restoring force. Fabricated suspended devices using chromium deposition, two levels of photolithography, Bosch deep silicon etching, hydrofluoric-acid release, and critical-point drying.
- Mounted released devices on a printed circuit board and used microscopy and metrology to characterize geometry and released structures before comparing voltage-dependent deflection and resonant-frequency measurements with theory.

Undergraduate Research

Investigation on the Nucleation and Growth of Palladium Nanoparticles by Electrodeposition Feb – Apr 2018

- Conducted undergraduate thesis research at the Indian Institute of Technology Bombay under Prof. Sankara Sharma V Tatiparti on the electrodeposition of non-spherical palladium particles.
- Modeled nucleation and particle growth during electrodeposition, modifying Scharifker–Hills models to represent instantaneous and progressive nucleation processes.

Effect of Copper Nanoparticles on FTO May – Jun 2017

- Investigated the effect of levetiracetam on copper nanoparticles deposited on fluorine-doped tin oxide (FTO) coated glass, using cyclic voltammetry to study the electrochemical response.
- Synthesized copper nanoparticles and characterized them using dynamic light scattering and UV–visible spectroscopy.

User Support and Teaching

Cleanroom Training and Laboratory Operations

Nanotech West, The Ohio State University | Training Lead and Cryogenic Probe Station Superuser

- Provided hands-on training to 10+ Nanotech West users in aligner, etcher, and metal-evaporator operation, supervising tool use and reinforcing operating protocols.
- Maintained and calibrated a Lakeshore cryogenic probe station and trained new users in measurement procedures.
- Mentored students across high-school, undergraduate, master's, and doctoral research in device fabrication, electrical measurements, and cleanroom protocols.

Cleanroom Project Assistant and Teaching Assistant Jun 2019 – Apr 2021

Quattrone Nanofabrication Facility, University of Pennsylvania

- Developed process travelers for ESE 336 and ESE 536 and standardized parameterized solar-cell fabrication recipes for cleanroom instruction. Designed and implemented the ESE 336 solar-cell fabrication module in Spring 2020.
- Supported hands-on cleanroom instruction in solar-cell fabrication and inkjet printing for undergraduate nanofabrication courses.
- Served as an assistant instructor for the August 2019 QNF Bootcamp, providing hands-on cleanroom tool training.
- Served as an ESE 336 teaching assistant from September 2019 to May 2020. Supported laboratory instruction and student questions, proctored quizzes, and managed course communications during the transition to hybrid learning.

Columbus School for Girls STEM Mentor Jun – Jul 2024

The Ohio State University

- Developed an accessible project on Al2O3 thin-film deposition using atomic layer deposition (ALD) and film analysis using ellipsometry, structuring the procedures so a high-school student could understand and follow them.
- Guided the student in understanding ALD and its semiconductor applications, with a focus on Al2O3 thin films.
- Provided hands-on explanations of ALD processes and techniques to encourage curiosity and develop technical understanding.
- Encouraged interest in advanced materials and semiconductor technology through one-on-one STEM mentoring.

Leadership and Professional Service

SPIE Student Chapter at Ohio State 2024 – 2026

Vice President

- Oversaw chapter operations and event logistics to support smooth functioning and alignment with SPIE's strategic goals.
- Built relationships with professors and academic leaders to organize seminars and technical talks that expanded learning opportunities for student members.
- Developed connections with industry professionals to support industry engagement and opportunities for student members.
- Coordinated with fellow student chapter leaders and SPIE representatives to support collaboration and a cohesive chapter environment.

Graduate Student and Postdoc Advisory Council 2024 – 2026

ECE Delegate

- Represented the interests and concerns of ECE graduate students, advocating for improvements to policies affecting research, community building, and campus well-being.

Research Mentorship

- Fritz Kyser, ECE graduate student, Ohio State (2025–2026).
- Jack Dvorak, ECE undergraduate student, Ohio State (2025–2026).
- Grace Campbell, Chris Chi, and Pranav Moola, ECE undergraduate capstone team, Ohio State (2025–2026).
- Naga Swetha Nallamothu and Sophie Mills, ECE graduate students, Ohio State (2024–2025).
- Ophelia Lonzo, Columbus School for Girls (2024).

Peer Reviewed Journal Articles

* Equal contribution.

1. N. S. Nallamothu, **M. Muduli**, Y. Xia, S. Lee, N. Gajowski, C. Chae, S. Arafin, S. Krishna, J. Hwang, R. M. Reano. GaAsSb–silicon interface analysis for photodetectors via direct wafer bonding. *Optical Materials Express*, 16(2), 309–320, 2026. doi:10.1364/OME.578886.
2. Y. Xia*, **M. Muduli***, S. Lee, N. Gajowski, S. Mills, N. S. Nallamothu, S. S. S. Nikor, R. L. Adams, R. M. Reano, S. A. Ringel, S. Krishna, S. Arafin. Investigation of the surface treatment of GaAsSb/Si using current– and capacitance–voltage measurements. *Journal of Applied Physics*, 139(17), 173102, 2026. doi:10.1063/5.0316959.
3. N. Gajowski, **M. Muduli**, T. J. Ronningen, S. Krishna. Impact ionization coefficients and excess noise characteristics of Al0.85Ga0.15As0.07Sb0.93 on GaSb substrate. *Applied Physics Letters*, 126(12), 2025. doi:10.1063/5.0258106.
4. Y. Xia, S. S. S. Nikor, N. Gajowski, S. Lee, **M. Muduli**, S. Noh, N. S. Nallamothu, R. L. Adams, R. M. Reano, S. A. Ringel, S. Krishna, S. Arafin. Dual-sacrificial-layer wet etching for transfer printing of InP-based antimony-containing materials. *ECS Journal of Solid State Science and Technology*, 14(10), 104004, 2025. doi:10.1149/2162-8777/ae10de.
5. **M. Muduli***, Y. Xia*, S. Lee, N. Gajowski, C. Chae, S. Rajan, J. Hwang, S. Arafin, S. Krishna. Active interface characteristics of heterogeneously integrated GaAsSb/Si photodiodes. *Applied Physics Letters*, 125(17), 2024. doi:10.1063/5.0230901.
6. H. N. Abbasi, S. Lee, H. Jung, N. Gajowski, Y. Lu, Y. Wang, D. Kim, J. Zhou, J. Gong, C. Chae, J. Hwang, **M. Muduli**, S. Nookala, Z. Ma, S. Krishna. Structural and electrical properties of grafted Si/GaAsSb heterojunction. *Applied Physics Letters*, 125(10), 2024. doi:10.1063/5.0225069.
7. J. Lee, T. Zhao, S. Yang, **M. Muduli**, C. B. Murray, C. R. Kagan. One-pot heat-up synthesis of short-wavelength infrared, colloidal InAs quantum dots. *The Journal of Chemical Physics*, 160(7), 2024. doi:10.1063/5.0187162.
8. M. Pise, **M. Muduli**, A. Chatterjee, B. P. Kashyap, R. N. Singh, S. S. V. Tatiparti. Instantaneous-Progressive nucleation and growth of palladium during electrodeposition. *Results in Surfaces and Interfaces*, 6, 100044, 2022. doi:10.1016/j.rsurfi.2022.100044.
9. T. Zhao, N. Oh, D. Jishkariani, M. Zhang, H. Wang, N. Li, J. D. Lee, C. Zeng, **M. Muduli**, H.-J. Choi, D. Su, C. B. Murray, C. R. Kagan. General Synthetic Route to High-Quality Colloidal III–V Semiconductor Quantum Dots Based on Pnictogen Chlorides. *Journal of the American Chemical Society*, 141(38), 15145–15152, 2019. doi:10.1021/jacs.9b06652.

Preprints

1. **M. Muduli**, N. Gajaowski, H. Jung, N. Nooman, B. Bhardwaj, M. Schwartz, S. Lee, S. Krishna. Lateral diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO. *arXiv*, 2024. doi:10.48550/arXiv.2409.20406.

Manuscripts Under Review

* Equal contribution.

1. **M. Muduli**, S. Mills, D. Plouffe, N. Gajowski, P. Murkute, Y. Xia, F. Kyser, S. Arafin, S. Krishna. Heterogeneously Integrated GaAsSb/Si Separate Absorption, Charge, and Multiplication (SACM) Photodiodes with Bonded Active Interface. *Research Square*, 2026. Under review. Preprint.
2. P. Murkute*, **M. Muduli***, N. Gajowski, Y. Khalifa, S. Krishna. Band Alignment and Interface Chemistry of Grafted GaAsSb/Si Heterojunctions. Under review.

Conference Proceedings

1. F. Kyser, **M. Muduli**, N. Gajowski, S. Krishna. Investigation of heterogeneous integration of GaAsSb with GaP/Si for SWIR applications. *Infrared Technology and Applications LII*, 14037, 2026. doi:10.1117/12.3097531.
2. Y. Xia, **M. Muduli**, N. S. Nallamothu, S. Mills, N. Gajowski, R. M. Reano, S. Krishna, S. Ringel, S. Arafin. Fabrication and Characterization of GaAsSb/Si Heterostructure Photodiode via Transfer Printing. *2025 IEEE Photonics Conference (IPC)*, 1–2, 2025. doi:10.1109/IPC65510.2025.11282307.
3. **M. Muduli**, N. Gajowski, Y. Xia, S. Lee, S. Arafin, S. Krishna. Heterogeneous integration of InGaAs/GaAsSb superlattices with Si for e-SWIR sensors. *Optical Interconnects and Packaging 2025*, 13372, 68–73, 2025. doi:10.1117/12.3040717.
4. N. S. Nallamothu, **M. Muduli**, Y. Xia, S. Lee, S. Arafin, S. Krishna, R. M. Reano. Heterogeneous integration of GaAsSb and silicon via direct wafer bonding for high-speed avalanche photodiodes. *Optical Interconnects and Packaging 2025*, 13372, 64–67, 2025. doi:10.1117/12.3042374.
5. M. Schwartz, H. Jung, S. Lee, **M. Muduli**, T. Ronningen, C. Ball, C. Grein, S. Krishna. Development of Antimonide Based Avalanche Photodiodes for SWIR Remote Sensing Applications. *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, 2023. doi:10.1109/IGARSS52108.2023.10283390.
6. **M. Muduli**, M. Schwartz, N. Gajowski, S. Lee, S. Krishna. Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO. *Infrared Technology and Applications XLIX*, 12534, 2023. doi:10.1117/12.2663602.

Book Chapters

* Equal contribution.

1. S. H. Kodati*, **M. Muduli***. Electrostatics in Engineering Photodiodes for Earth Observation Cameras. *Electrostatics - Fundamentals and Modern Applications*, 2025. doi:10.5772/intechopen.1009104.
2. P. P. Das*, **M. Muduli***, S. Borah, V. Chaudhary. Principle of Green Chemistry: A modern perspective for development of sustainable textile fiber-based green nanocomposites. *Green Chemistry for Sustainable Textiles*, 121–136, 2021. doi:10.1016/B978-0-323-85204-3.00031-2.

Conference Abstract

1. C. Ball, S. Krishna, T. J. Ronningen, S. Lee, H. Jung, M. Schwartz, **M. Muduli**, C. Grein, E. Fuller. Development of Avalanche Photodiodes with High Operating Temperatures to Enable Short-Wavelength Infrared Remote Sensing Applications. *AGU Fall Meeting*, Abstract A15L-1383, 2022.

Technical Expertise

Process & materials integration: Wafer bonding · 3D heterogeneous integration · micro-transfer printing · 300 mm wafer processing · interface engineering

Semiconductor fabrication: Photolithography · e-beam lithography · ALD · CVD · PVD · RIE · wet/dry etching · metallization · RTA

Materials & device characterization: XPS · SEM · AFM · ellipsometry · profilometry · SIMS · IV/CV · QE · PL · UV–Vis and optical spectroscopy

Simulation & data analysis: Silvaco TCAD · NovaTCAD · MATLAB · JMP · DOE · SRIM · Origin · Layout Editor · KLayout · Lumerical

Nanomaterials: InAs, InSb, InAsSb, and CdSe quantum dots; PbS, Ag, Au, Pd, and Cu nanoparticles

Tools and laboratory methods: Mask and maskless aligners, spin coating, e-beam evaporation, plasma ashing, cryogenic probing, reflectometry, optical microscopy, and dynamic light scattering.

Oral Presentations

SPIE Photonics West 2025
San Francisco, CA
- Heterogeneous Integration of InGaAs/GaAsSb Superlattices with Si for e-SWIR Sensors.

Electronic Materials Conference 2024
College Park, MD
- GaAsSb/Si Heterojunction Photodiodes Fabricated with Epitaxial Layer Transfer.

SPIE Defense + Commercial Sensing 2023
Orlando, FL
- Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO.

Poster Presentations

Heterogeneous Integration of Antimonide–Semiconductors with Si for Infrared Sensors September 26, 2025
Kraus Poster Competition · The Ohio State University, Columbus, OH

Heterogeneous Integration of Antimonide–Semiconductors with Si for Infrared Sensors February 26, 2025
Hayes Forum · The Ohio State University, Columbus, OH

Heterogeneously Integrated GaAsSb/Si Photodiodes with an Active Interface September 11, 2024
8th US–Japan Digital Innovation Hub and Advanced Technology Workshop · Columbus, OH

Heterogeneously Integrated GaAsSb/Si Photodiodes with an Active Interface September 7, 2024
Kraus Poster Competition · The Ohio State University, Columbus, OH

Demonstration of Zn-Diffusion in 2-µm InGaAs/GaAsSb Superlattice Planar Diodes using Atomic Layer Deposition of ZnO November 15, 2023
IEEE Poster Presentation · The Ohio State University, Columbus, OH

Demonstration of Zn-Diffusion in 2-µm InGaAs/GaAsSb Superlattice Planar Diodes using Atomic Layer Deposition of ZnO September 13, 2023
Kraus Poster Competition · The Ohio State University, Columbus, OH

Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO May 4, 2023
Materials and Manufacturing Conference · The Ohio State University, Columbus, OH

Awards and Recognition

2025 | **Hayes Forum Poster Competition:** Third place, The Ohio State University.

2025 | **SPIE Student Travel Award:** For SPIE Photonics West 2025, San Francisco; selected in November 2024.

2024 | **Electronic Materials Conference:** Honorable Mention, Best Student Paper Award.

2024 | **Dr. Burn Lin Travel Grant Award:** Recipient.

2024 | **Beam of Light Award:** KIND Laboratory, The Ohio State University; recognized for a collaborative approach to research.

2023 | **Kraus Student Poster Competition:** Second place, The Ohio State University.

2023 | **Dr. Burn Lin Travel Grant Award:** Recipient.

2022 | **Make OHI/O:** First place for a team-developed 3D-printed Braille generator.

2018 | **Shri Baljit Shastri Award:** Amity University, Noida; Best in Human and Traditional Values.